



Introduction to Machine Learning

“CAN MACHINES THINK?”

-- ALAN TURING

What is intelligence?



What makes humans (*Homo sapient*) distinct from animals?
Language, writing, reasoning, creative invention, moral choice ...

What is intelligence?

- ▶ Intelligence is the ability to **think** and understand instead of doing things by instinct or automatically.

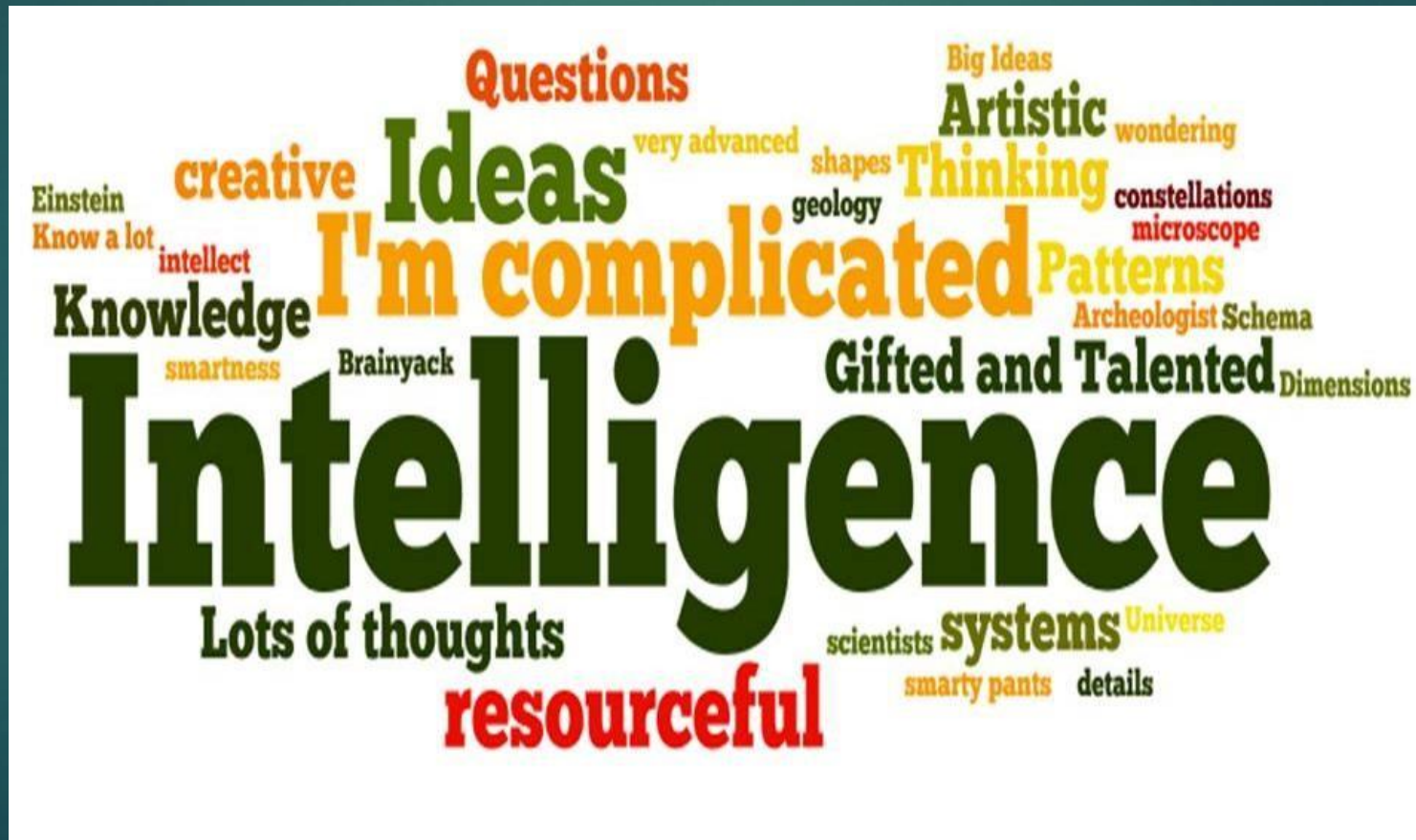
(Essential English Dictionary, Collins, London, 1990)

- ▶ But, What “thinking” means?
- ▶ Thinking is the activity of using your brain to consider a problem or to create an idea.

(Essential English Dictionary, Collins, London, 1990)

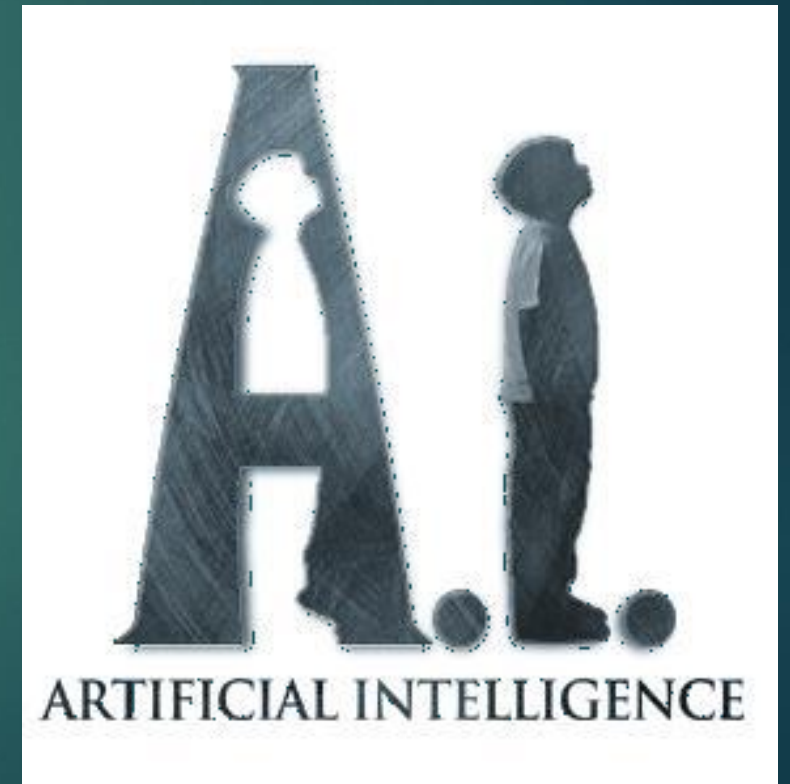
What is intelligence?

- ▶ Intelligence is the ability to learn and understand, to solve problems and to make decisions.



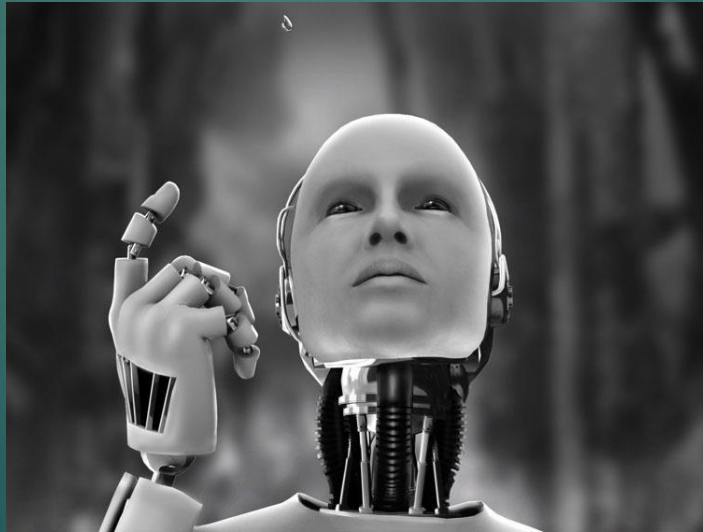
Human vs Non-Human

- ▶ Can non-humans (man-made objects) have Intelligence?"
- ▶ To be more specific, "Can Machines Think?"
- ▶ To address this question, a new discipline was born ...



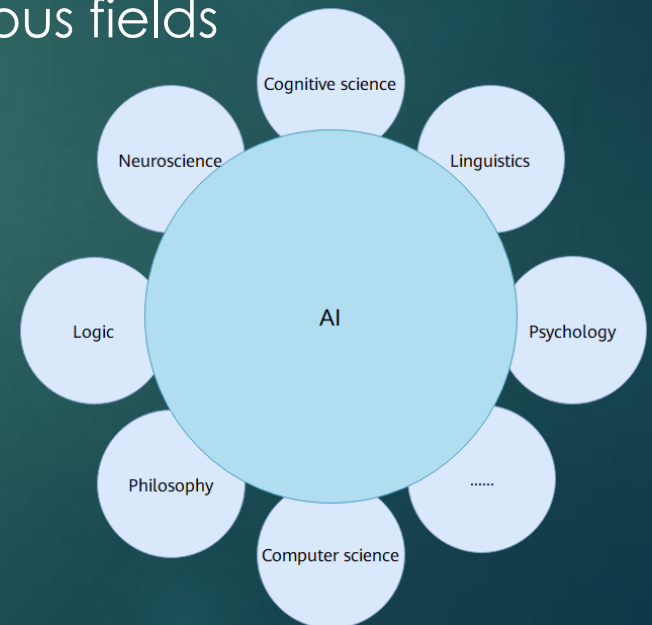
Artificial Intelligence

- ▶ “A science to make machines do things that would require intelligence* if done by humans.”



Some Refresh about Artificial Intelligence

- ▶ "science and engineering of making intelligent machines, especially intelligent computer programs". ---John McCarthy in 1956
- ▶ The very premise of AI technology is to enable machines to learn from collected data, and make human-like decisions.
- ▶ Thus, AI is a scientific discipline that studies and develops theories, techniques, and application systems used to simulate and extend human intelligence.
- ▶ AI has become an interdisciplinary course that involves various fields

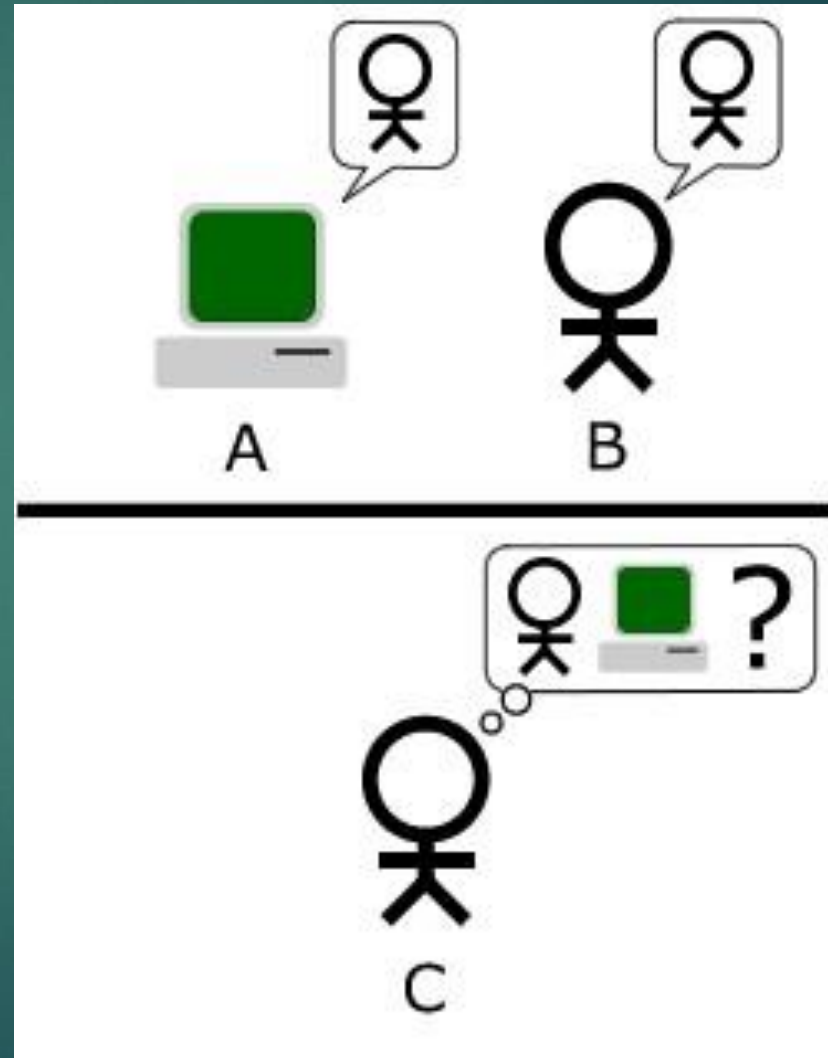


Some Refresh about AI

- ▶ The Turing Test by Alan Turing
- ▶ The main purpose: to test the machine's ability to exhibit intelligent behaviour that is indistinguishable from that of a human
- ▶ Instead of asking 'can a machine think', Turing was asking: Can a machine pass a behavioral test for intelligence?
- ▶ Turing focused on **observable behavior**—if a machine can convincingly mimic human intelligence, it should be considered intelligent

Turing Test

- ▶ Is it a machine or a human?
- ▶ Test Setup:
 - ▶ A human judge interacts with a machine and a human through text-based communication.
 - ▶ If the judge cannot reliably distinguish the machine from the human, the machine is said to have passed the test.

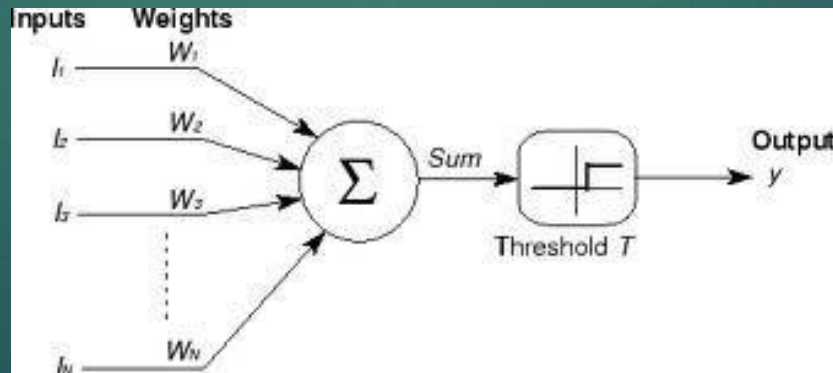


Three Important Era of AI (Early Day)

- ▶ The era of great expectations (1956 – late 1960s)
- ▶ Unfulfilled promises (late 1960s – early 1970s)
- ▶ The key to success (early 1970s – mid-1980s)

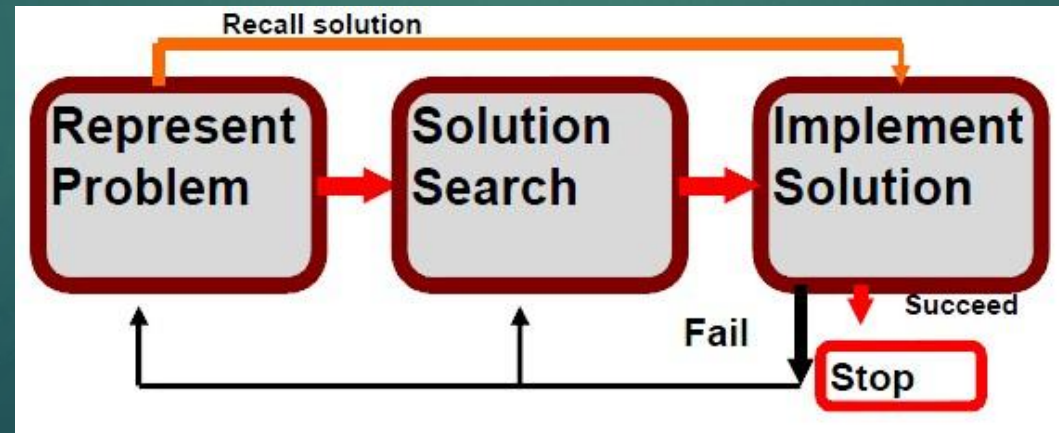
The era of great expectations (1956 – late 1960s)

- ▶ The early works on perceptron started by McCulloch and Pitts (1957) was continued.
- ▶ Learning methods were improved and Frank Rosenblatt (1958) proved the perceptron convergence theorem, demonstrating that his learning algorithm could adjust the connection strengths of a perceptron.



General Problem Solver

- ▶ One of the most ambitious projects was the General Problem Solver (GPS)
 - ▶ To simulate the thinking process by inventing **general methods** for solving broad classes of problems
- ▶ **Key idea:** To solve problems, programs applied a search strategy by trying out different combinations of small steps, until the right one was found.



General Problem Solver

- The methods were based on **formal logic** and could generate an infinite number of possible operators – **Symbolic AI**.

DL	FOL
$a : C$	$C(a)$
$\langle a, b \rangle : P$	$P(a, b)$
$C \subseteq D$	$\forall x. C(x) \rightarrow D(x)$
$P^+ \subseteq P$	$\forall x, y, z. (P(x, y) \wedge P(y, z)) \rightarrow P(x, z)$
$\top \subseteq \leq 1 P$	$\forall x, y, z. (P(x, y) \wedge P(x, z)) \rightarrow y = z$
$P \equiv Q^-$	$\forall x, y. P(x, y) \iff Q(y, x)$
$C_1 \sqcap \dots \sqcap C_n$	$C_1(x) \wedge \dots \wedge C_n(x)$
$C_1 \sqcup \dots \sqcup C_n$	$C_1(x) \vee \dots \vee C_n(x)$
$\neg C$	$\neg C(x)$
$\{a_1, \dots, a_n\}$	$x = a_1 \vee \dots \vee x = a_n$
$\exists P.C$	$\exists y. (P(x, y) \wedge C(y))$
$\forall P.C$	$\forall y. (P(x, y) \rightarrow C(y))$
$\geq n P.C$	$\exists y_1, \dots, y_n. \bigwedge_{1 \leq i \leq n} (P(x, y_i) \wedge C(y_i))$
$\leq (n-1) P.C$	$\forall y_1, \dots, y_n. (\bigwedge_{1 \leq i < n, i < j \leq n} y_i \neq y_j \rightarrow (\bigwedge_{1 \leq i \leq n} (P(x, y_i) \wedge C(y_i))))$

- This approach was quite feasible for simple toy problems (eg. Hanoi Tower Problem) at that time

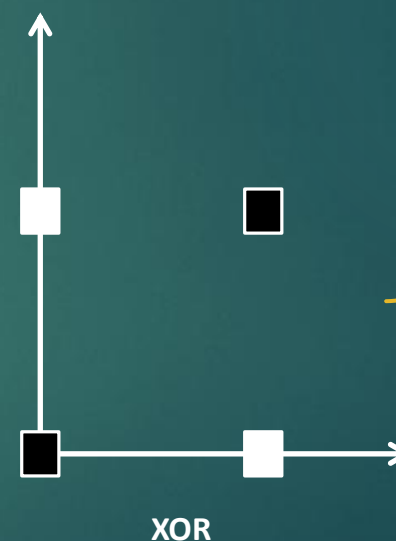
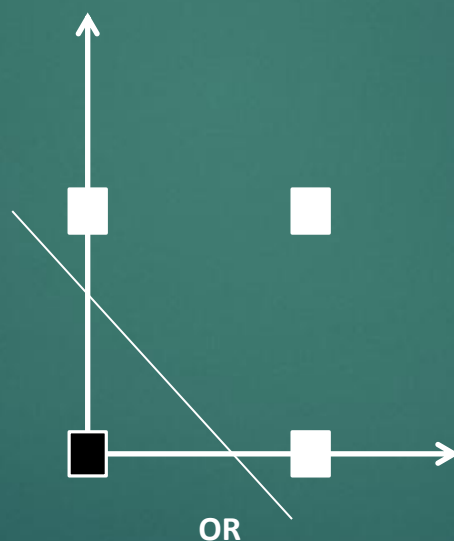
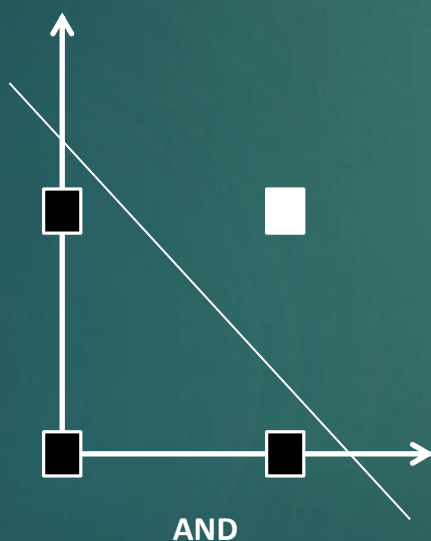
General Problem Solver

- ▶ Naturally, people expected that If the programs could be “scaled up” to solve large problems, they would finally succeed
- ▶ However, GPS failed to solve complex problems
- ▶ The amount of computer time and memory that GPS required to solve real-world problems led to the project being abandoned

Unfulfilled Promises (late 1960s – early 1970s)

► XOR Problem

- In 1969, Marvin Minsky, an American mathematician and AI pioneer, proved that perceptrons are essentially a type of linear model capable of processing only linear classification and cannot solve XOR (exclusive OR) problem



Inputs		Output
A	B	Z
0	0	0
0	1	1
1	0	1
1	1	0

- By 1970, the euphoria about AI was gone, and most government funding for AI projects was cancelled

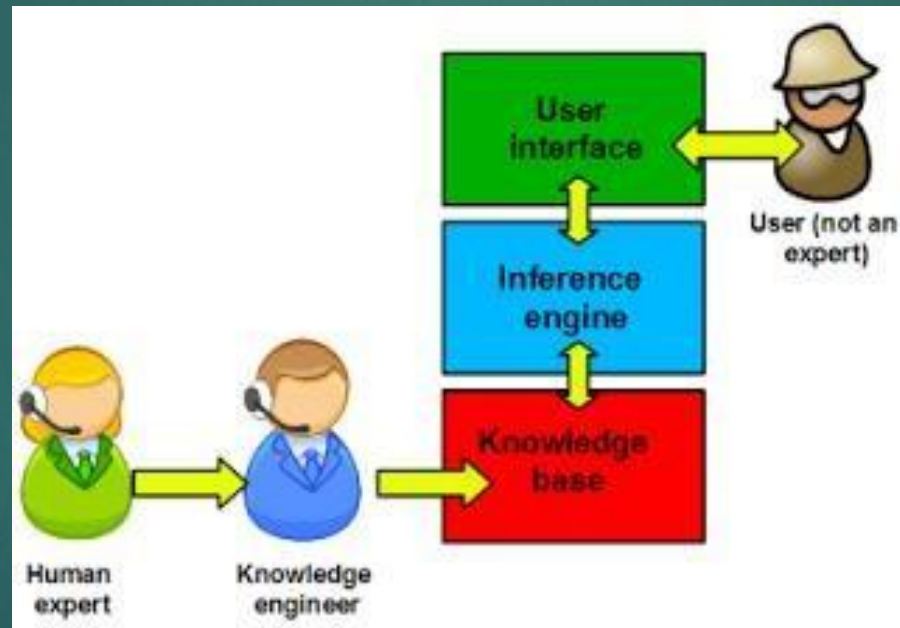
Unfulfilled Promises (late 1960s – early 1970s)

- ▶ WHY?

- ▶ AI was still a relatively new field, academic in nature, with few practical applications apart from playing games.
- ▶ **NONE** of the AI system at that time could manage real-world problems.

The key to success (early 1970s – mid-1980s)

- ▶ When weak methods failed, researchers finally realised that the only way



- ▶ to deliver practical results was to solve typical cases in **narrow areas of expertise+making large reasoning steps** – the birth of **Expert Systems**.

Expert System – MYCIN (Stanford University)

- ▶ MYCIN (1970s) was a rule-based expert system for the diagnosis of infectious blood diseases.
- ▶ MYCIN's knowledge consisted of ~ 600 rules derived from human knowledge in a narrow domain through extensive interviewing of experts

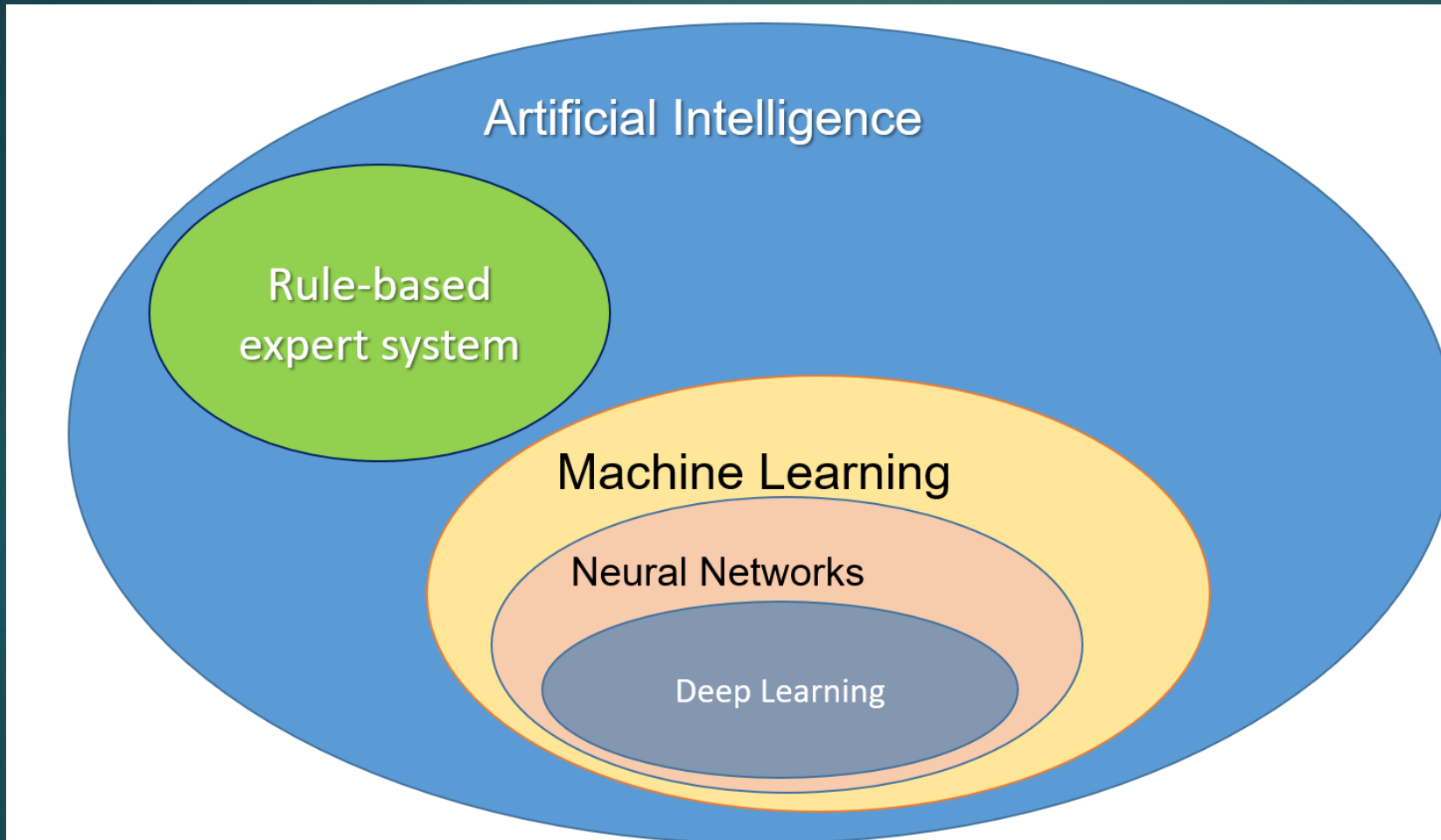
However

- ▶ Expert systems are restricted to a very narrow domain of expertise
- ▶ Expert systems, especially the 1st generation, have little or no ability to learn from their experience
- ▶ Expert systems are built individually and cannot be developed fast. Complex systems can take over 30 person-years to build

Data-driven AI = Computational AI

- ▶ Decline of Symbolic AI, rising of Computational AI (CI)
- ▶ Symbolic AI – symbols, tools – formal logic
- ▶ Example: GPS, automated theorem proving, Knowledge Representation
- ▶ Computational AI – data, tools – Statistics, biological inspired techniques, Math optimization etc

Overview of AI, ML and DL



Relationship Between AI, Machine Learning, and Deep Learning

AI Four basic elements: data, algorithms, computing power, and application scenarios

Machine learning

Deep learning

1950

1980

2010

Strong vs Weak AI

- ▶ Weak AI (artificial narrow intelligence)
 - ▶ Specialize in one area
 - ▶ Expert Systems, Intelligence Personal Assistant (Siri/Cortana/Google Home/Alexa etc), AlphaGo and many others
- ▶ Strong AI (artificial general intelligence)
 - ▶ A computer that is as smart as a human *across the board*—a machine that can perform any intellectual task that a human being can